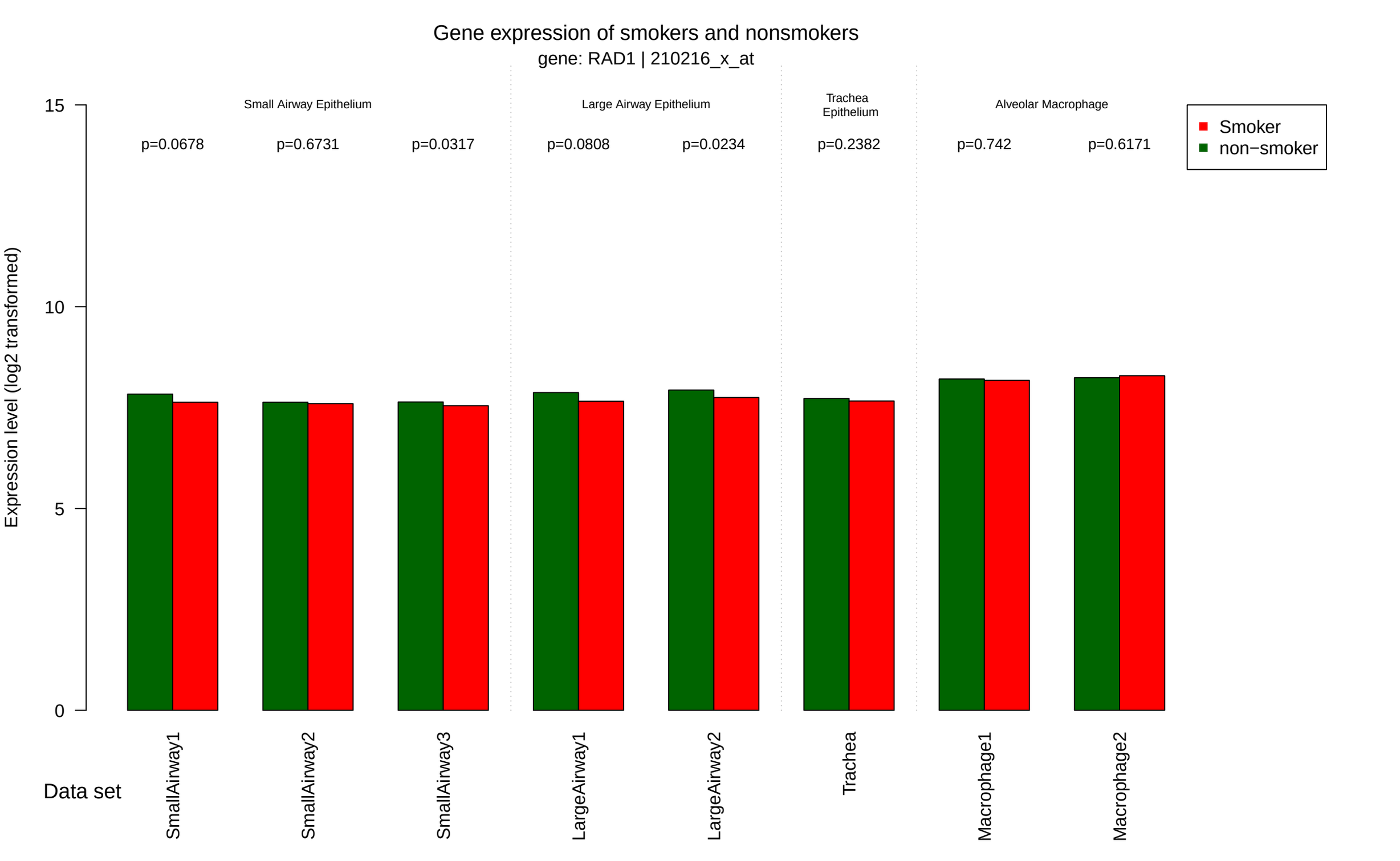
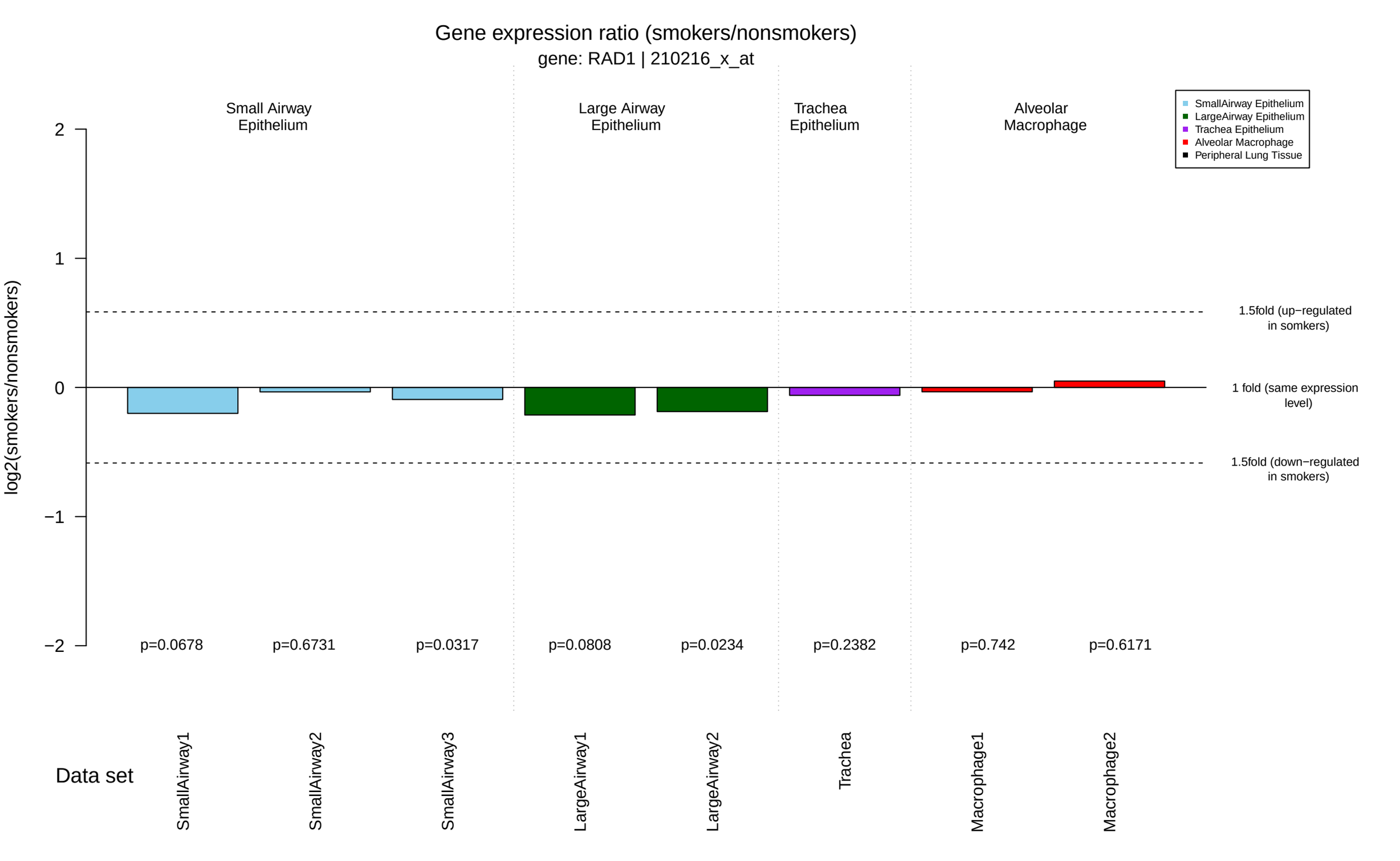
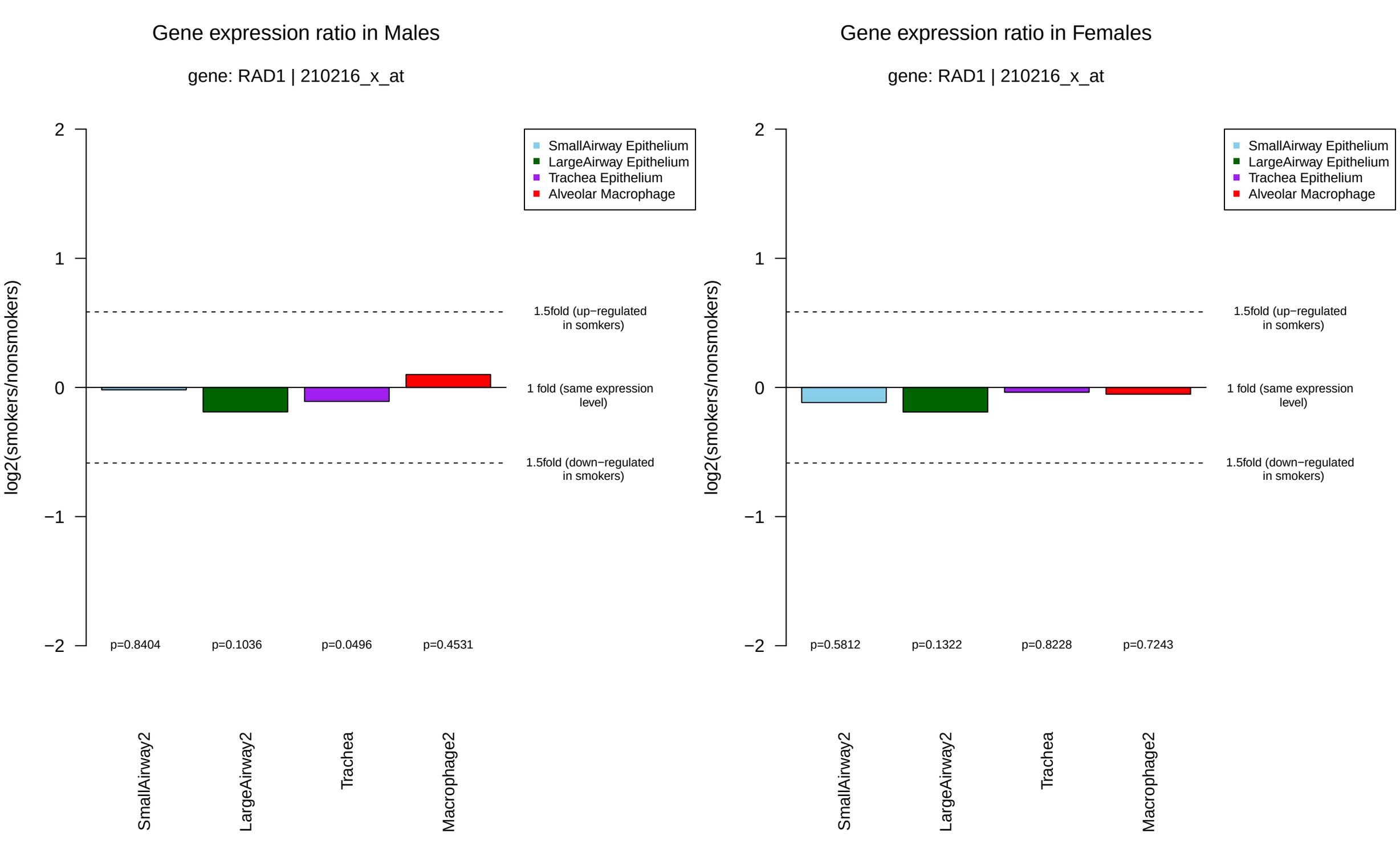
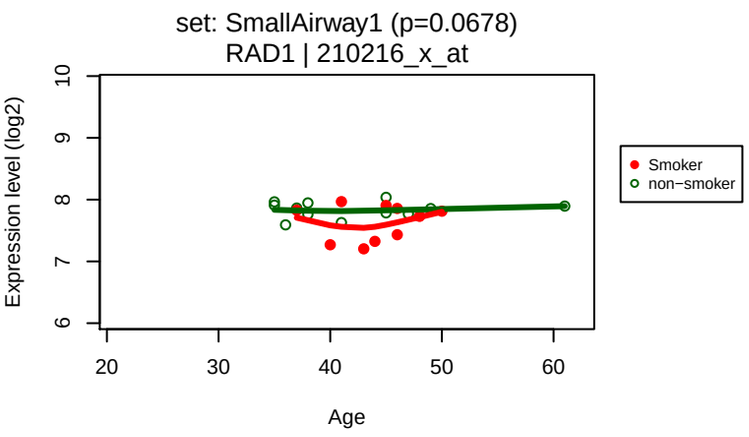
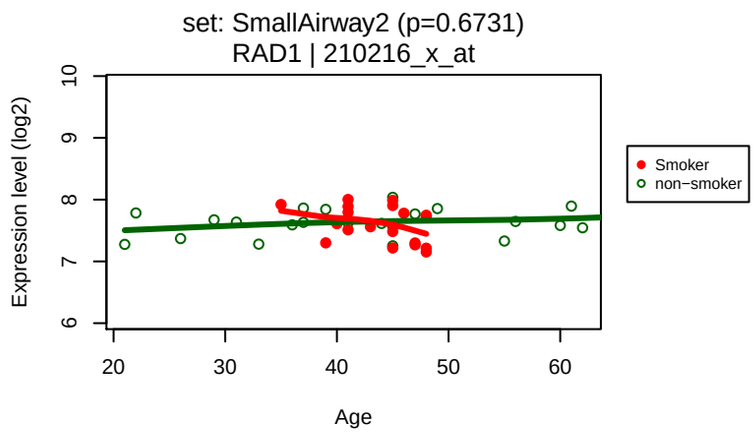
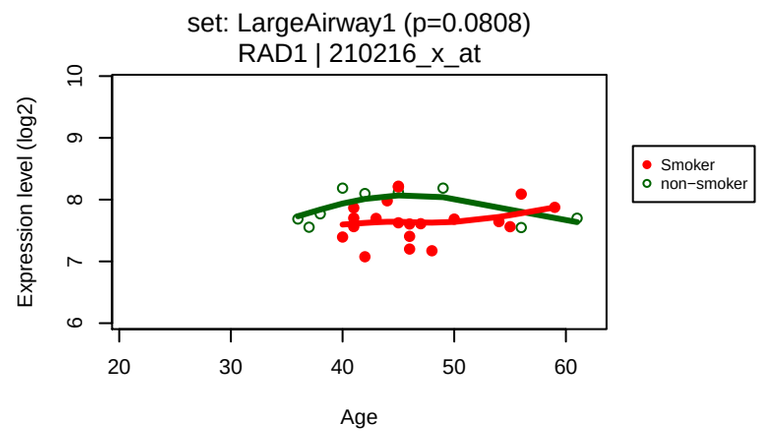

Gene expression of smokers and nonsmokers
gene: RAD1 | 210216_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0678
p=0.6731
p=0.0317
p=0.0808
p=0.0234
p=0.2382
p=0.742
p=0.6171
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

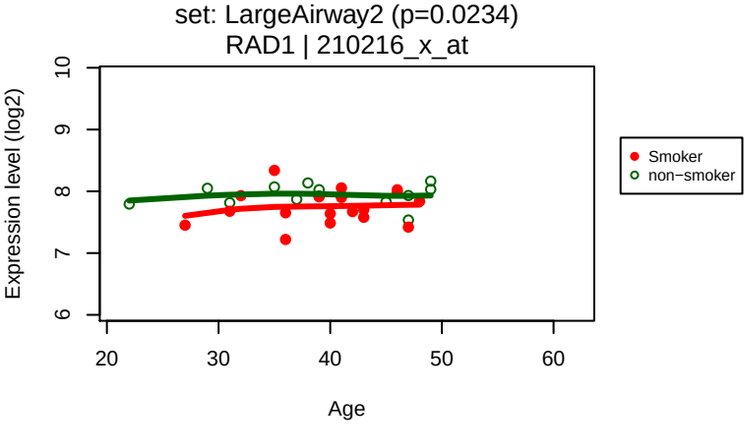

Gene expression ratio (smokers/nonsmokers)
gene: RAD1 | 210216_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0678
p=0.6731
p=0.0317
p=0.0808
p=0.0234
p=0.2382
p=0.742
p=0.6171
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

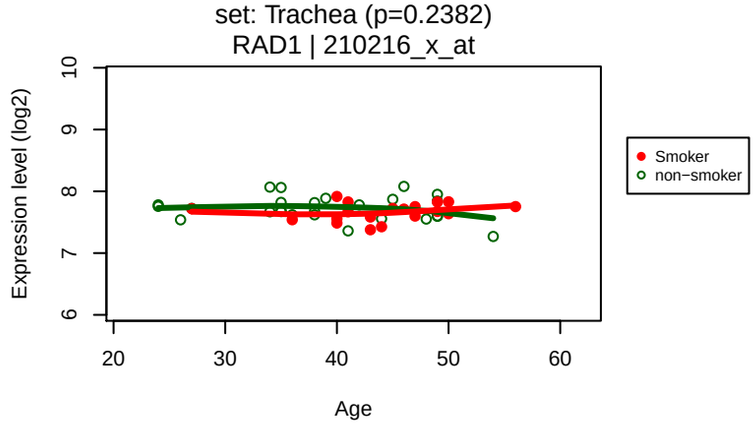

Gene expression ratio in Males
gene: RAD1 | 210216_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8404
p=0.1036
p=0.0496
p=0.4531
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD1 | 210216_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5812
p=0.1322
p=0.8228
p=0.7243
SmallAirway2
LargeAirway2
Trachea
Macrophage2

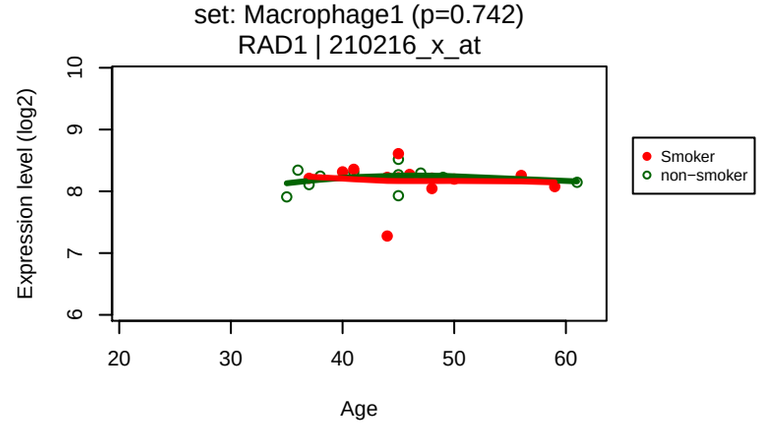

set: SmallAirway1 (p=0.0678)
RAD1 | 210216_x_at
Expression level (log2)
Age
Smoker
non−smoker

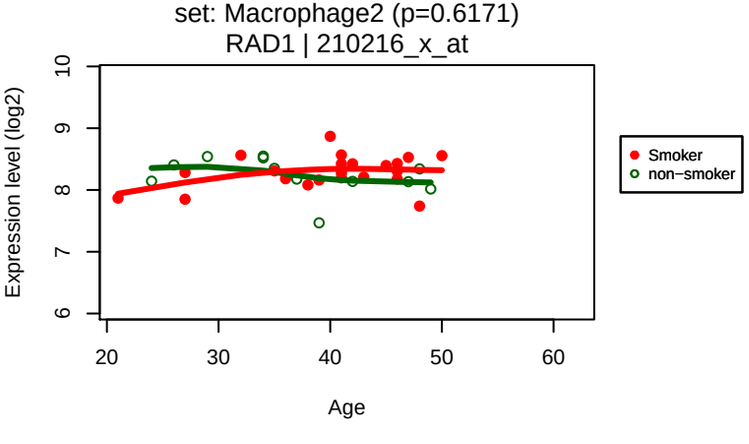

set: SmallAirway2 (p=0.6731)
RAD1 | 210216_x_at
Expression level (log2)
Age
Smoker
non−smoker

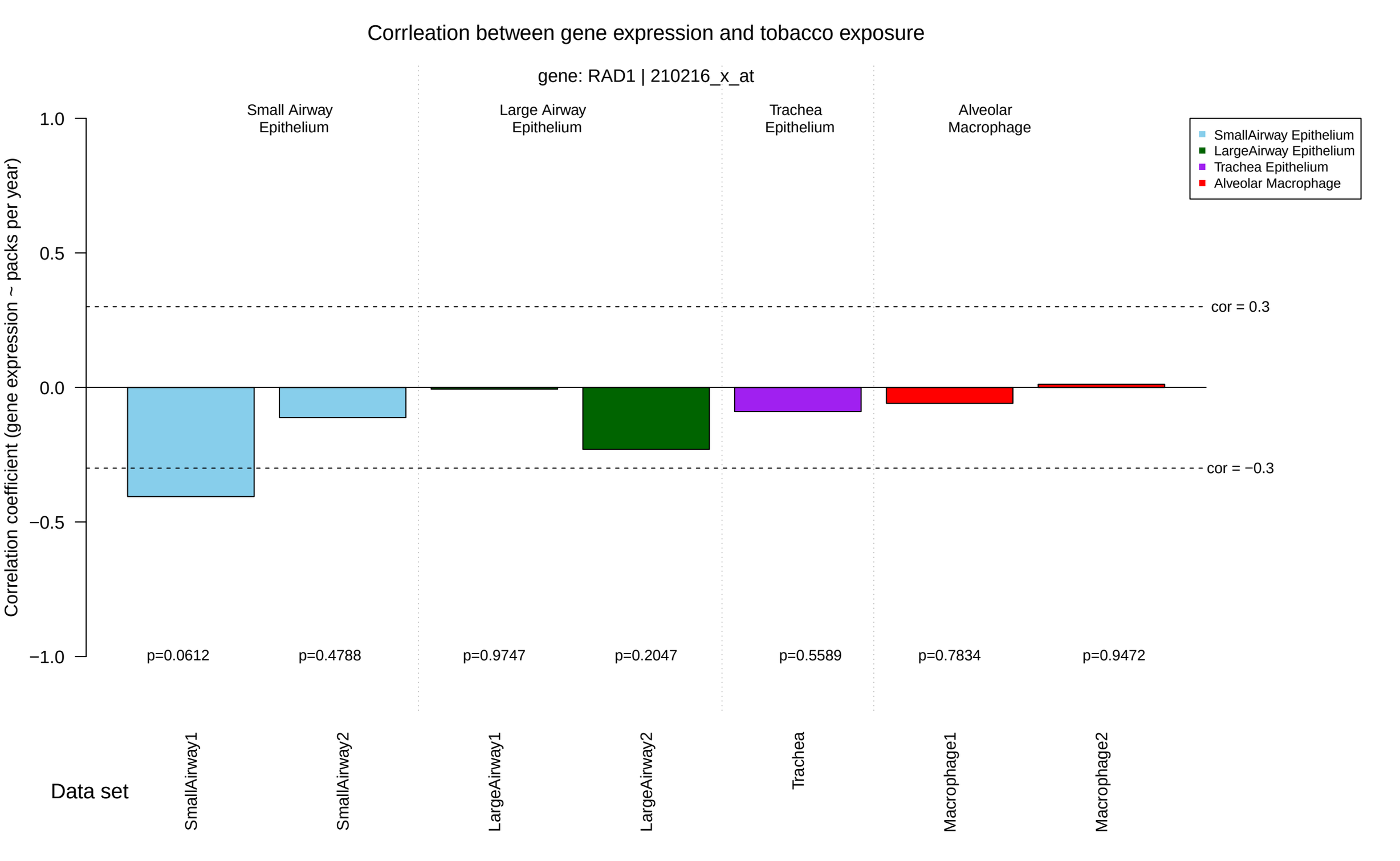

set: LargeAirway1 (p=0.0808)
RAD1 | 210216_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0234)
RAD1 | 210216_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2382)
RAD1 | 210216_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.742)
RAD1 | 210216_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6171)
RAD1 | 210216_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD1 | 210216_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0612
p=0.4788
p=0.9747
p=0.2047
p=0.5589
p=0.7834
p=0.9472
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2